Response to the Associate Editor and Reviewers

Manuscript 26-0591v1: Informativity and Identifiability for Identification of Networks of Dynamical Systems

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We thank the associate editor and the three reviewers for their careful assessment. The revised technical note distinguishes data informativity from network identifiability, states the admissible model class more precisely, provides proofs of the four theorems, and supplies a reproducible MATLAB supplement. References to sections, theorems, and examples below refer to the revised manuscript. The submitted manuscript's references [3], [17], and [19] are, respectively, the papers by Hendrickx, Gevers, and Bazanella; van der Woude; and Legat and Hendrickx. We address the comments individually below.

Response to the Associate Editor

Comment.

> *Three reviewers evaluated the manuscript, and they converge on a set of substantive concerns regarding novelty, rigor, and presentation, which are summarized below:*

Response. We recognize the editor's assessment and the need for a major revision before resubmission. The revised note narrows its contribution claims, states its assumptions, and gives proofs and worked calculations for the results retained below.

AE.1: Novelty and positioning

Comment.

> *1. Novelty and positioning of the results — Reviewer 5; Reviewer 4. The main results appear to be straightforward extensions of well-known results. In particular, Reviewer 5 notes that the link between vertex-disjoint paths and the rank of associated matrices (Theorem 2) is already established in reference [3] of the manuscript. Reviewer 4 similarly observes that Theorem 2 "seems like a correct generalization" of a result in [17] but that the extension is not shown to be nontrivial, and asks whether Theorem 3 is new. The paper does not clearly delineate what is original versus what is a restatement of existing work, which must be clarified.*

Response. The revised abstract identifies the broader model class and the two-solution example. Three bullets at the end of Section I now state explicitly what is novel relative to the cited work:

the scope of the joint analysis, its model-specific rank tests, and the identifiability analysis for the broader model with repeated and constrained entries. They identify the spectral, path-rank, and fiber-dimension principles as established results. Section IV applies the path result to $\Pi = RP^{-1}Q + S$ under independent free entries and gives a separate Gröbner basis rank test for known or constrained entries. Section V applies the finite-fiber criterion to the closed-loop map $F(X) = R_y(X)P(X)^{-1}Q_r(X) + S_{yr}(X)$, allowing unknown input and measurement transfer functions and repeated entries. Example 3 illustrates the analysis by computing two solutions that a generic Jacobian-rank test does not count.

AE.2: Proofs and references

Comment.

> *2. Missing proofs or references for the main theorems — Reviewer 3 ; Reviewer 4. Theorems 1–3 are given without formal proofs or precise references. Reviewer 3 notes that the informativity results are presented only informally as generalizations of prior work, and recommends inserting formal proofs stating exactly which assumptions are necessary. Reviewer 4 points out that Theorem 1 is loosely stated (the spectrum should be positive definite "for almost all \omega"), and that references on the "transfer of excitation" problem (e.g., Gevers, Bazanella, Bombois, Miskovic, IEEE TAC, Vol 54, No 12) should be provided for context.*

Response. The revised manuscript includes a proof of Theorem 1 using the external-signal spectrum and the rank of Π, a determinant and path proof for Theorem 2, an elimination proof for Theorem 3 with the nonzero-denominator restriction, and a fiber-dimension proof for Theorem 4. The paper cites the underlying informativity, path-rank, and algebraic-geometry results. Theorem 1 states the network, well-posedness, spectral, predictor-stability, excitation, and generic-rank assumptions explicitly.

AE.3: Concepts and terminology

Comment.

> *3. Conceptual clarity and terminology — Reviewer 4; Reviewer 5; Reviewer 3. The manuscript mixes the notions of identifiability, network identifiability, generic/local/global identifiability without clear distinction. Reviewer 4 notes that the definition offered in the introduction conflates classical identifiability with network identifiability, and that the claim that partial measurement was not previously considered is incorrect. Reviewer 5 stresses that informativity and identifiability are distinct concepts that deserve separate treatment, and flags that Example 3 appears to contradict a conjecture discussed in [19], which should be carefully verified.*

Response. Sections I and V distinguish predictor informativity, generic network identifiability, and generic local network identifiability in the algebraic sense. The introduction now says explicitly that the latter permits several compatible networks, and Section V explains that the criterion does not certify an exceptional realization. Example 3 is no longer presented as a counterexample to the Legat–Hendrickx question: it uses unknown input and measurement transfer functions and a repeated unknown entry, which lie outside that paper's model class.

AE.4: Model and assumptions

Comment.

> *4. Model formulation and assumptions — Reviewer 5; Reviewer 3; Reviewer 5. The motivation for the model P(q)wk =Q(q)uk , zk =R(q)wk +S(q)uk is unclear: Reviewer 5 asks why two matrices P(q) and Q(q) are needed and notes the unusual separation of state and output dynamics compared to standard network models. Reviewer 3 points out that invertibility of P is assumed without stating conditions or references, that it is not stated whether the network topology is known, and that it is unclear how to select which predictors to use. Reviewer 5 further notes that the function F(X)=Ry P-1Qr +Syr is written as a function of X, yet X does not appear in the expression.*

Response. Section II embeds the familiar $w = Gw + Bu$, $z = Cw$ model as $P = I - G$, $Q = B$, $R = C$, $S = 0$. It now motivates the separate P and Q through physical networks with local node dynamics, coupling laws, and a separate input matrix, citing Kivits and Van den Hof. It also gives a discrete-time descriptor model with $P(q) = qE - A$ as an illustration of a coefficient-constrained structure. Normalizing the node equations recovers an $I - G$ form but introduces shared unknown factors. Filtered or combined measurements and direct input effects motivate R and S. Section II also states the known structure, separates generic invertibility from well-posedness, and checks $\det P$ in the examples. Section III explains how measured signals define the predictor.

AE.5: Presentation and readability

Comment.

> *5. Presentation and readability — Reviewer 5; Reviewer 4. Several sections (Section VI on sub-networks, and the numerical examples in Sections VII–VIII) are perceived as adding little and could be removed or substantially condensed. The paper relies on concepts from algebraic geometry (affine varieties, ideals, fibers, Grobner bases) that are not standard in the controls community and should be clearly introduced. The symbol z is used for two different quantities, and there are numerous typos (e.g., publication year "2913" in reference [21]).*

Response. We condensed the active subnetwork discussion, revised the examples around distinct uses of the rank and fiber criteria, and added short explanations of the algebraic terminology at its point of use. Section VIII explains when path, Jacobian-rank, and elimination tests are useful and notes the cost of symbolic elimination. The computational examples give software and hardware details and point to `supplementary_matlab_code.zip`. The shift operator is q, the complex argument is λ, and the output signal is z_k.

Comment.

> *Based on the collective reviews and this editor's own reading, this editor finds that the manuscript addresses a relevant topic and the reviewers acknowledge potential value in some ideas. However, the manuscript in its current form lacks formal justification for its main theorems, does not adequately position its novelty relative to the established literature, and contains unresolved conceptual and modeling ambiguities. These concerns*

go beyond minor revision, as the reviewers question whether the core results are nontrivial extensions. Based on these grounds, this editor suggest to reject this manuscript but with the option that it could be resubmitted as a technical note after major revision. If the authors decide to resubmit, the authors need to address all concerns raised by the reviewers in the revision, and moreover, the authors are reminded that IEEE TAC has the policy that the manuscript will be rejected if it cannot be accepted with minor revision in the next round of review.

Response. We have revised the manuscript in response to the editor and all three reviewers. The detailed changes are described in the replies below.

Response to Reviewer 3

Comment.

This manuscript deals with the informativity and identifiability conditions for network of dynamic systems. The authors focus their analysis using the so-called Grobner basis and provide graph-theoretic conditions. For informatitvity, sufficient conditions are derived in terms of the rank of a matrix of rational functions, relating the excitation signals to the inputs of the optimal predictor. The identifiability is investigated under the lens of the dimensionality of a proper fiber for the local identifiability. The paper assumes full excitation of the network, but partial measurement. This paper also provides an example under which it states that local identifiability does not imply generic identifiability, a hypothesis that was previously discussed in the literature. This paper is clear and well-written and there is also a clear contribution to the field, but it is hard to follow specially because the authors tend to omit key steps. Therefore, I have some concerns about the results of the paper given in the following.

Response. We thank the reviewer for identifying where assumptions and steps needed to be stated explicitly.

R3.1: Invertibility and well-posedness of P

Comment.

1. The network model used in this manuscript assumes that P is invertible for well-posedness. No remarks nor references are given to state under what conditions this transfer matrix is invertible.

Response. Section II now assumes $\det P$ is not identically zero as a rational expression in the free entries. This is a case-by-case test for generic invertibility. A particular realization is called well-posed only when its rational matrix P is invertible; then $P(\lambda)$ is invertible except at finitely many complex arguments where it is defined. The examples display the relevant determinants. Stability is imposed separately where needed.

R3.2: Formal proofs and necessary assumptions

Comment.

> *2. There are no formal proofs for none of the Theorems presented in the informativity section. The ideas behind the proofs are simple given in informal terms, and mostly they are cited as generalizations of works from other. I recommend inserting formal proofs for all the results stating exactly which are necessary, instead of just referring to a book covering a wide topic.*

Response. Theorems 1–3 in Section IV now have proofs. Theorem 1 requires a positive-definite external-signal spectrum for almost all frequencies and full generic row rank of Π; it concludes informativity for algebraically generic admissible true networks, not for every realization. It also states well-posedness of the true network, existence of the external-signal spectrum, and stability of the admissible candidate predictors. Theorem 2 permits prescribed nonzero diagonal entries of P and requires the other structurally nonzero entries to be independent free variables. It uses a Schur-complement determinant identity to relate the rank of the transfer function matrix to vertex-disjoint paths. Theorem 3 defines the denominator restriction $D \neq 0$, the minor ideal and the auxiliary equation $1 - tD = 0$, then proves the three rank cases by elimination. A remark after Theorem 2 also covers fixed selector matrices Q, R, formed from distinct columns and rows of the identity matrix, respectively, with independent free structurally nonzero entries in S. More general known and constrained entries are handled by the algebraic rank test. Example 1 contains fixed off-diagonal entries of P, so its informativity conclusion follows from the displayed determinant of Π, rather than an application of Theorem 2.

R3.3: Choice of predictor

Comment.

> *3. While the characterization is given in terms of input to predictors input, how exactly one knows which predictors to use? There are several ways to select the predictors.*

Response. Section III makes the choice of measured output y_k and measured input r_k explicit through $z_k = (y_k, r_k)$. Under the assumed stable innovation representation, this choice defines the predictor and the closed-loop transfer function G_c that it determines. Different choices lead to different G_c and require their own informativity test. The predictor-reduction remark in Section IV and Example 2 show how known relations among predictor columns can permit a smaller measured-input vector. The input matrix Q need not excite every node. We do not claim a universal optimal predictor-selection rule.

R3.4: Known topology

Comment.

> *4. Does one need to assume that the topology of the network is known? This assumption is not stated in the manuscript.*

Response. Section II now states that the structural zero pattern, known entries, and specified relations among unknown entries of (P, Q, R, S) are given. The analysis determines unknown entries within this model class; topology recovery is outside its scope. Theorem 2 requires independent free entries except for prescribed nonzero diagonal entries of P; its following remark also permits fixed selector matrices Q, R. The Gröbner basis procedure allows more general prescribed entries and specified relations.

R3.5: Constructing the rank-test Gröbner basis

Comment.

> *5. Theorem 3 seems interesting by its own, but it is not clear how to construct the Gronber basis from it.*

Response. Section IV gives the construction: clear row denominators to obtain $\widetilde{\Pi}$, adjoin all k-minors and $1 - tD$ to the ideal, place t first in a lexicographic order, and inspect basis polynomials independent of t on $D \neq 0$. Example 3 works through the full-row-rank case: its reduced basis is $\{t-1, H\}$, so rank drops exactly when $H = 0$. The auxiliary variable excludes singular or otherwise undefined realizations.

R3.6: Constructing the identifiability Gröbner basis

Comment.

> *6. The same comment applies to Theorem 4, which derives a necessary and sufficient condition for local identifiability.*

Response. Section V describes how to clear denominators in $F(X) = G_c$, adjoin $1 - tD_F$, eliminate X and t, and compute the dimension of the resulting image closure. Example 3 shows the polynomial equations, a lexicographic variable order, the zero elimination ideal, and the resulting image dimension. It also solves the fiber equation to show why equal dimensions can coexist with two admissible networks.

Response to Reviewer 4

Comment.

> *The paper treats the identification of networks, particularly the problem of their identifiability, which is an important topic and has been receiving particular attention from the community in the past 15 years or so. It brings new and relevant ideas to the field and the results seems correct. Yet, there are various issues to be resolved before publication in these Transactions can be considered.*

Response. We thank the reviewer for the detailed comments on terminology, prior work, and proofs.

R4.1: Terminology for identifiability

Comment.

> *In the introduction the concepts of informativity and identifiability are briefly presented. These are rather confused, as "identifiability" is a classical concept in the identification literature that usually refers to parametrizations of transfer functions, whereas the definition offered in the introduction ("whether the network topology is such that the open loop transfer functions can be uniquely determined from the predictor.") looks more like what is called "network identifiability" in the literature of network identification. In the introduction itself the authors "correct" themselves in this regard when, in the last paragraph, they talk about local network identifiability instead. Please clarify.*

Response. The introduction now separates identification of a predictor from recovery of the unknown open-loop transfer functions. The latter is called *network identifiability*, distinct from identifiability of a numerical parameterization. Section V defines generic local network identifiability as a zero-dimensional generic algebraic fiber, which allows finitely many solutions; generic network identifiability requires uniqueness. This finite-ambiguity meaning now appears in the introduction as well. The manuscript also distinguishes this symbolic rational-function formulation from the single-frequency local notion in Legat and Hendrickx.

R4.2: Earlier work on partial measurements

Comment.

> *It is also said that more general networks are considered than in previous works, which is correct. But It is also said "we will consider the partial measurement case . . . " implying that previous works all considered the full measurement case, which is far from the truth. Please correct.*

Response. The introduction now cites prior work on partial node measurements and subnetworks. We do not present partial measurement alone as new. The model permits unknown input and measurement transfer functions, specified relations among unknown entries, and a direct input-to-output term; these are the aspects compared with prior work in Sections I, II, and V.

R4.3: Standard MIMO informativity and Definition 1

Comment.

> *Regarding informativity, what is studied is if the input signals are rich enough to identify the "closed-loop" relationship from u to z. So, this is standard MIMO identification, there is nothing "networkly" here, is it? Then definition 1 seems the standard definition in the literature, so some reference to it would be welcome; if this is not the standard definition, references are still needed, and the difference(s) should be commented. By the way, this is also the case in other parts of the paper, where some known results or definitions are stated without references; this too must be improved.*

Response. We agree that the positive-definite predictor-input spectrum is a standard informativity condition, for which we now cite Ljung's *System Identification: Theory for the User*. Definition 1 is attributed to the prediction-error informativity formulation of Bombois et al.; the discussion also cites Gevers, Bazanella, Bombois, and Miskovic for network-specific excitation. The manuscript explains why predictor uniqueness is the relevant data condition before asking whether open-loop transfer functions are uniquely recovered. The network-specific step is computing the generic rank of $\Pi = RP^{-1}Q+S$ from the known interconnection structure or from specified algebraic constraints. Informativity and network identifiability are developed in separate sections.

R4.4: Distinct uses of z

Comment.

> *One should not use 'z' to denote two different things.*

Response. The measured output signal retains the conventional notation z_k. Transfer functions use q as the shift operator and λ as a complex argument. The revised manuscript no longer uses z as a second transform variable.

R4.5: Theorem 1 and transfer of excitation

Comment.

> *Now, the main issue: the Theorems. Theorems 1 to 3 are given without proof or reference! In Theorem 1 the statement itself is loose: it should probably read "the spectrum is positive definite for almost all omega . . . ". With this statement the Theorem is probably true, but it remains to be proven. Also, references on this "transfer of excitation" problem should be given for context and comparison, so that we (the readers) can tell whether the result is new; for that, I suggest that the authors look at the following paper from these Transactions and some of the references therein: Gevers, Bazanella, Bombois, Miskovic - Identification and the information matrix: how to get just sufficiently rich? Vol 54, No 12.*

Response. Theorem 1 now explicitly requires the external-signal spectrum to be positive definite for almost all frequencies, meaning outside a set of Lebesgue measure zero. Its proof uses $\Phi_z = \Pi\Phi_u\Pi^*$ and full row rank of Π for algebraically generic admissible true networks. The citation to Gevers et al. provides the requested transfer-of-excitation context, while Ljung is cited for the classical spectral informativity implication. The manuscript presents that implication as standard rather than a new network theorem. The theorem also specifies well-posedness, existence of the external-signal spectrum, and stability of the candidate predictors.

R4.6: Theorem 2 and the earlier path result

Comment.

Theorem 2 indeed seems like a correct generalization of the very nice result in [17], but the extension is not trivial. Also, more context is welcome here, as similar disjoint paths conditions have been extensively used in the field.

Response. We cite van der Woude's structured-system result. Theorem 2 allows prescribed nonzero diagonal entries of P, while all other structurally nonzero entries must be independent free variables. The revised proof uses $\det\begin{bmatrix} P & Q_J \\ -R_I & S_{I,J} \end{bmatrix} = \det(P)\det(\Pi_{I,J})$. Determinant terms give disjoint input-output paths, including one-edge paths through S. It now explains how the permutation terms split into paths, cycles, and diagonal factors. The entries selected outside the diagonal of P determine the remaining diagonal factors, so prescribed nonzero diagonal entries do not introduce cancellation. A remark defines selector matrices Q, R and explains why the determinant argument also applies to them. Here S may be zero or have independent free entries. For $P = I - G$ and $S = 0$, this recovers the path-rank result for selected submatrices of $(I-G)^{-1}$ in Hendrickx, Gevers, and Bazanella. We present the general path/rank principle as prior work and specify how the present matrix form is covered.

R4.7: Theorem 3, references, and novelty

Comment.

Theorem 3 also needs a proof and/or references to provide better context. Is this Theorem new? Overall, the proofs of the Theorems - or references to the proofs if they are not original - must be given and more reference in the text to the papers in the reference list (which may be amended a little) must be used to provide better context.

Response. Theorem 3 now has an elimination proof and an explicit ideal construction; Example 3 gives a small calculation. We use standard Gröbner basis elimination and cite the algebraic-geometry references for that method. The claim is the rank test for this model with its admissible denominator domain and allowed constraints, not a new general theorem about Gröbner bases. The revised manuscript also explains the role of the auxiliary variable t and attributes the fiber-dimension step in Theorem 4 to algebraic geometry.

Response to Reviewer 5

Comment.

This paper presents an extension of well-known results on identifiability and informativity to a more general networked model. This reviewer has several comments:

Response. We thank the reviewer for the detailed assessment of the model, results, and examples.

R5.1: Motivation for P, Q, R, S

Comment.

1.- The motivation of this new model is unclear. The authors propose a model of the form $P(q)w_k = Q(q)u_k$ $Z_k = R(q)w_k + S(q)u_k$. It is not justified why there are two matrices P(q) and Q(q) instead of a single matrix. In addition, it is unusual that the dynamics for the state and the output are different. Typically, the models in network identification are of the form: $w_k = G(q)w_k + Br_k + v_k$ $y_k = Cw_k$, Natural extensions could involve considering matrices of transfer functions in B and/or C. However, a comparison or justification of the proposed model is not provided.

Response. Section II now displays the standard network $w_k = G(q)w_k + B(q)u_k$, $z_k = C(q)w_k$ as the special case $P = I - G$, $Q = B$, $R = C$, $S = 0$. The roles of the four matrices are stated there: P describes internal connections, Q how inputs enter the nodes, R how nodes are measured, and S direct input-to-output paths. The model permits unknown entries in Q and R, direct paths through S, and known relations among entries of P, Q, R, S. Local physical dynamics and couplings define the node equations while actuators define a separate input matrix, as in the physical-network model of Kivits and Van den Hof cited in Section II. Normalizing the node equations can yield an $I-G$ form, but the resulting transfer functions share factors and constraints, which the constrained-entry test retains. Section II also gives a discrete-time descriptor example with $P(q) = qE-A$, $Q = B$, $R = C$, and $S = D$. For singular E, this form retains algebraic equations and makes prescribed zero patterns and known coefficients explicit. Identifiability of its real matrix coefficients requires retaining their constant-coefficient structure; the example motivates the model form without claiming a separate descriptor identifiability result. Filtered sensors and direct input effects motivate R and S; Example 3 uses repeated unknowns across matrices.

R5.2–3: Novelty of the main results and the path/rank link

Comment.

2.- The main results appear to be just straightforward extensions of well-known results in informativity and identifiability of linear networks. That is why there are no mathematical proofs, except for Theorem 4, which is a bit trivial. 3.- Theorem 2 related to the link of vertex-disjoint paths with the rank of associated matrices is already well-known in identifiability of linear networks from the reference [3] of the manuscript.

Response. We agree that the disjoint-path principle is not new; the submitted reference [3] studies network identifiability with partial node measurements, and [17] gives the structured-system rank result. The revised abstract states the model class, and three bullets at the end of Section I identify the novel model-specific contributions relative to the cited work. Section IV cites van der Woude's structured-system result and the submitted reference [3] as related network path literature. It explains that the paths characterize the rank of the full closed-loop transfer function matrix in the P, Q, R, S model, including direct paths through S. Theorem 2 records the direct-feedthrough extension for the present $RP^{-1}Q + S$ form, allowing prescribed nonzero diagonal entries of P. The following remark defines fixed selector matrices Q, R and relates their path-rank condition with $P = I - G$, $S = 0$ to the submitted reference [3]. It also permits independent free entries in S. Theorem 3 handles more general known and constrained entries through polynomial ideals. Section V then applies a dimension test to the closed-loop map with unknown input and measurement transfer functions. We have confined the novelty claim to this model class and its combined rank and fiber analysis. Example 1 uses a direct determinant calculation because its fixed off-diagonal entries of P fall outside the graph theorem's assumptions.

R5.4: Separation of informativity and identifiability

Comment.

> *4.- Informativity and identifiability are usually two different concepts as it has been stated in the reference [13] of the manuscript. Analyzing both subjects simultaneously does not seem natural and I think they deserve separate treatment as is usually done.*

Response. The introduction now describes the two questions separately. Sections IV and V treat them in order: informativity concerns whether measured signals distinguish candidate predictors; network identifiability concerns which open-loop transfer functions are determined by the closed-loop transfer function G_c. The latter step can have a finite ambiguity even when the informativity condition holds, as Example 3 shows.

R5.5: Length and role of the examples

Comment.

> *5.- Several sections of the paper do not provide new information and give the impression that they could be easily removed. There is a full page of only examples corresponding to Section VII. In addition, Section VIII Numerical Examples does not provide interesting information. Even some of the comments are rather vague like we can compute in 11 minutes on a laptop computer .*

Response. We revised the examples around distinct points of the analysis. Example 1 separates generic uniqueness from an exceptional zero-edge realization; Example 2 shows predictor-input reduction from known structure; Example 3 shows a two-point algebraic fiber and a worked Gröbner basis calculation. The five-node calculations report image dimensions and generic rank for a larger example. We replaced the vague runtime description with the measured computation time, software versions, processor and memory, and prepared the MATLAB supplement with both entry scripts, three helper files, and run instructions. The computational section now states when the path and Jacobian-rank tests may avoid elimination and cautions that the reported runtime is not a scaling law.

R5.6: Subnetwork section

Comment.

> *6.- The Section VI Sub-networks could be easily removed by analyzing from the beginning the case of sub-networks where the full network case can be easily encompassed.*

Response. We retained a shorter separate section because the boundary signals measured for a subnetwork affect the predictor inputs and the informativity calculation in the full network. Section VI writes both direct boundary measurements and measured combinations of boundary signals in the P, Q, R, S notation. This makes clear which transfer function is identified for each selection without introducing a second model class.

R5.7: Example 3 and the Legat–Hendrickx question

Comment.

> *7.- The authors seem to mix notions of identifiability, generic identifiability, local identifiability, global identifiability, etc. That is why in Example 3, they seem to contradict the conjecture in the reference [19] of the manuscript. The authors should verify this because according to the authors in [19], the conjecture has already been proven up to five unknown transfer functions in [19]. Also, in Example 3, the authors claim that there is always another pair of matrices that satisfy the identifiability. Does this imply just that the network is unidentifiable?*

Response. We no longer present Example 3 as a counterexample to the question in submitted reference [19]. That paper considers fixed node-selection matrices B, C and independent unknown entries of G. Our Example 3 has unknown entries in Q_r and R_y, with the same A appearing in both. In our model its generic fiber has exactly two points: A and $A' = -HG - A$, with G and H determined by G_c. For proper stable G, H, A, the alternative A' is proper and stable. Thus the example is generically locally identifiable in the algebraic sense but not generically uniquely network identifiable within our broader model class. It does not resolve the question for the narrower Legat–Hendrickx class.

R5.8: Visible dependence of F on X

Comment.

> *8.- The authors introduce the function F of the form $F(X) = R_y P^{-1} Q_r + S_{yr}$. This should be a function of X, but it does not appear in the function. What is X?*

Response. Section V defines X as the vector of free entries of (P, Q_r, R_y, S_{yr}) and writes

$$F(X) = R_y(X)P(X)^{-1}Q_r(X) + S_{yr}(X).$$

The subsequent discussion keeps λ symbolic so equality means equality of rational transfer functions.

R5.9: Zero edge in Example 1

Comment.

> *9.- In Example 1, the authors mention that they do not have global network identifiability because there is one solution only when $G_c^1 \neq 0$. By definition, this case $G_c^1 \neq 0$ is excluded since it implies that there is no edge or connection.*

Response. Section II now distinguishes the specified structural zero pattern from the value of a free entry at a particular realization. A structurally free entry may vanish exceptionally. Example 1 explicitly states that $G^1 = 0$ is permitted but nongeneric, and that uniqueness fails there while generic network identifiability holds.

R5.10: Algebraic terminology

Comment.

> *10.- The authors use concepts like affine varieties, ideals, fibers, Grobner basis, etc. that are not standard in the control systems community and for this reason, these concepts should be clearly introduced in the manuscript to improve readability.*

Response. The revised text explains the ideal generated by polynomials, polynomial zero sets (varieties), irreducibility, dimension, the Zariski closure, fibers, and elimination where each is used. Section IV explains how a lexicographic Gröbner basis isolates the rank-deficient set. Section V and Example 3 show the corresponding image and fiber calculations.

R5.11: Typos and references

Comment.

> *11.- There are many typos in the paper like the publication year (2913) in the reference [21].*

Response. We corrected notation, grammar, and terminology throughout the manuscript. The cited bibliography year is 2013. The revised version uses z_k for the measured output signal, q for the shift operator, and λ for a complex argument. We also corrected Dankers's name in the bibliography and consolidated two citations to different editions of the same algebraic-geometry book.

Materials accompanying the revision

The source manuscript is `main.tex`, and the prepared MATLAB archive is `supplementary_matlab_code.zip`. The archive contains the two five-node example scripts, their three helper functions, and a README specifying the commands and software dependencies. The manuscript points to this archive in Section VIII.